Date:14/10/26

Experiment No. 1

Generation and Classification of Continuous-Time and Discrete-Time Signals

OBJECTIVE:

(i) To generate and visualize basic continuous-time signals such as unit step, unit impulse, ramp, sinusoidal, and exponential signals.

(ii) To generate and visualize basic discrete-time signals and analyze their amplitude and time-index characteristics .

SOFTWARE / LABORATORY ENVIRONMENT:

MATLAB

I. To generate and visualize basic continuous-time signals such as unit step, unit impulse, ramp, sinusoidal, and exponential signals.

PROCEDURE:

1. Define a continuous time vector with a sufficiently small simulation interval.

2. Generate unit impulse, unit step, ramp, exponential and sinusoidal signals.

3. Plot each waveform against time using suitable amplitude and time scales.

4. Measure amplitude, frequency and period wherever applicable.

5. Compare the simulated waveform with its theoretical expression.

6. Record observations and justify the signal characteristics.

MATLAB CODE:

```
clc;
clear;
close all;


% Time axis
t = -2:0.001:2;


% Changed signal parameters
A = 3;
```

```
f0 = 10;
a = 2;


% Generation of signals
imp = double(abs(t)<0.0005);
step = double(t>=0);
ramp = t.*step;
expSig = exp(-a*t).*step;
sinSig = A*sin(2*pi*f0*t);


% Plotting all five signals
figure;


subplot(3,2,1);
plot(t,imp);
grid on;
title('Unit Impulse');
xlabel('Time (s)');
ylabel('Amplitude');


subplot(3,2,2);
plot(t,step);
grid on;
title('Unit Step');
xlabel('Time (s)');
ylabel('Amplitude');

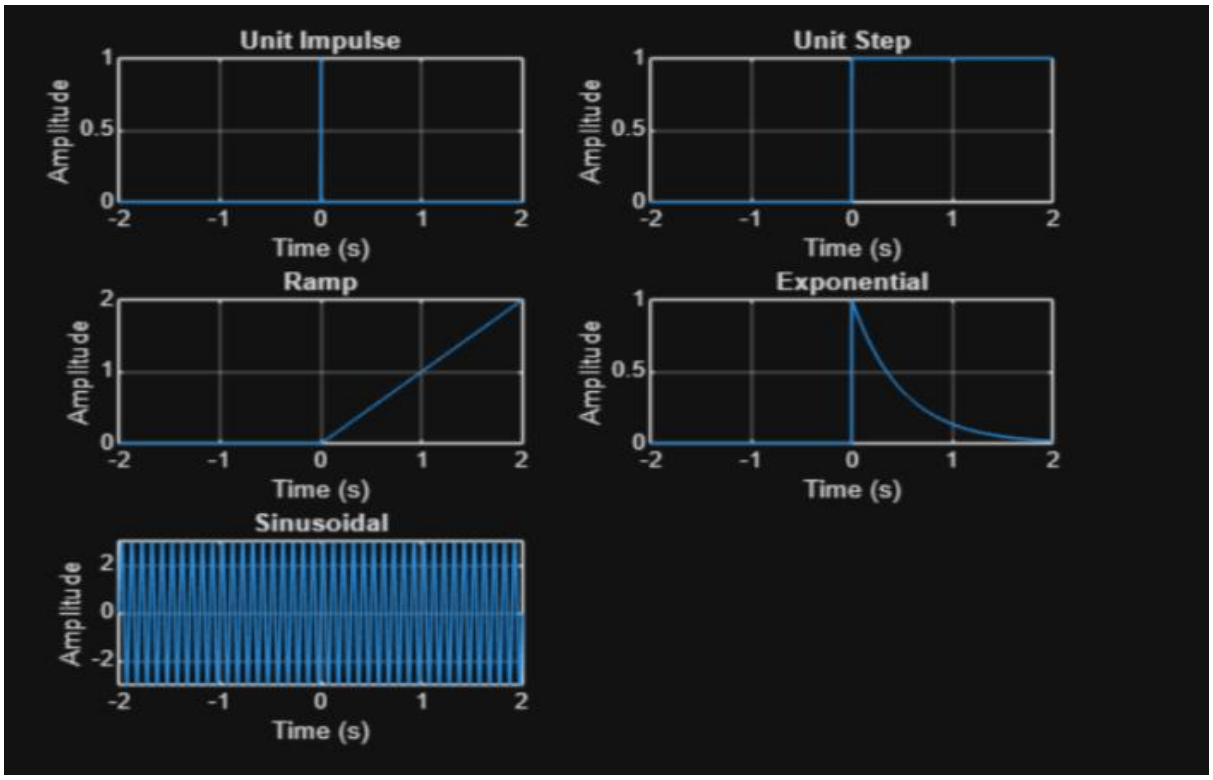

subplot(3,2,3);
plot(t,ramp);
grid on;
title('Ramp');
xlabel('Time (s)');
ylabel('Amplitude');


subplot(3,2,4);
plot(t,expSig);
grid on;
title('Exponential');
xlabel('Time (s)');
ylabel('Amplitude');


subplot(3,2,5);
plot(t,sinSig);
grid on;
title('Sinusoidal');
```

```
xlabel('Time (s)');
ylabel('Amplitude');
```

OUTPUT:



II. To generate and visualize basic discrete-time signals and analyze their amplitude and time-index characteristics .

PROCEDURE:

1. Define the integer sample index n.

2. Generate unit impulse, unit step, ramp, exponential and sinusoidal sequences.

3. Display each sequence using a stem plot.

4. Determine sample amplitudes and period where applicable.

5. Classify the sequence as periodic/non-periodic and even/odd wherever applicable.

6. Compare the generated samples with theoretical values and record observations.

MATLAB CODE:

```
clc;
clear;
close all;


% Time index interval
n = -25:25;


% Amplitude and frequency values
A = 3;
f0 = 0.08;
a = 0.15;


% Signal generation
imp = double(n == 0);
step = double(n >= 0);
ramp = n .* step;
expSig = (0.85).^n .* step;
sinSig = A * sin(2*pi*f0*n);


% Plotting all signals
figure;


subplot(3,2,1);
stem(n, imp, 'filled');
grid on;
title('Unit Impulse');


subplot(3,2,2);
stem(n, step, 'filled');
grid on;
title('Unit Step');

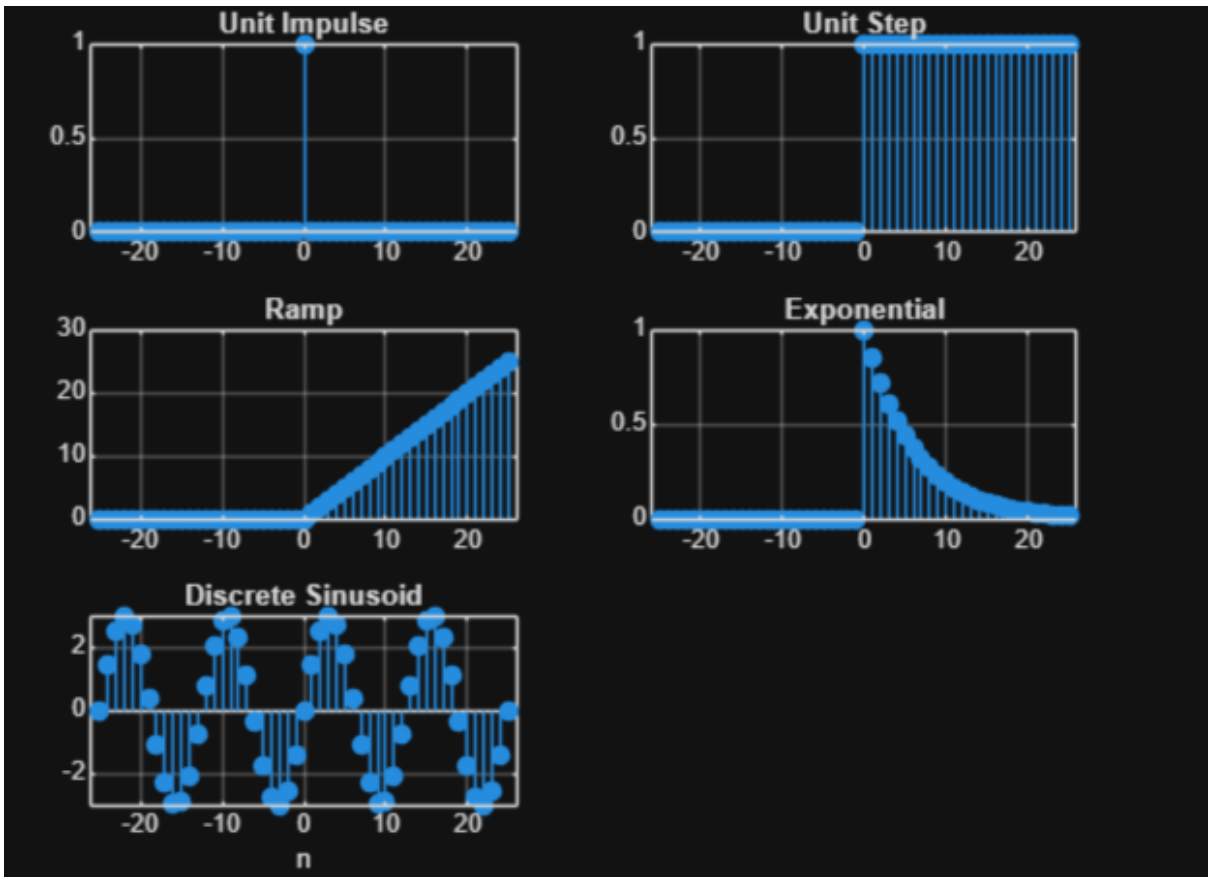

subplot(3,2,3);
stem(n, ramp, 'filled');
grid on;
title('Ramp');


subplot(3,2,4);
stem(n, expSig, 'filled');
grid on;
title('Exponential');
```

```
subplot(3,2,5);
stem(n, sinSig, 'filled');
grid on;
title('Discrete Sinusoid');


xlabel('n');
```

OUTPUT:



RESULT:

The continuous-time and discrete-time signals were generated and analyzed successfully .

LABORATORY PERFORMANCE ASSESSMENT RUBRICS:

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage	Marks(10)	Marks obtained
1. Generate and visualize basic continuous-time signals such as unit step, unit impulse, ramp, sinusoidal and exponential.	**CT Signal Generation & MATLAB Implementation**	Generates all five CT signals accurately with a suitable time vector and step size. Uses correct MATLAB commands . Plots are correctly scaled, with titles, axis labels and grid.	Generates the required CT signals and plots correctly, with minor errors in code, time range, parameters or labelling.	Generates basic CT signals but has noticeable errors in code, step or impulse representation, or plots, and needs assistance.	Unable to generate the required CT signals or produce meaningful plots.	**25**	**2.5**	
2. Generate and visualize basic discrete-time signals such as unit sample, unit step, ramp, sinusoidal and exponential sequences.	**DT Signal Generation & MATLAB Implementation**	Generates all five DT sequences accurately with a proper index vector n. Uses stem correctly. Plots are correctly scaled, with titles, axis labels and grid.	Generates the required DT sequences and plots correctly, with minor errors in index range, stem usage, parameters or labelling.	Generates basic DT sequences but has noticeable errors in code, indexing or stem plots, and needs assistance.	Unable to generate the required DT sequences or produce meaningful stem plots.	**25**	**2.5**	
3. Classify CT and DT signals as periodic or aperiodic and identify their waveform characteristics.	**Signal Classification & Periodicity Analysis**	Correctly classifies all signals using DT. Determines the period where applicable. Identifies frequency,	Classifies most signals correctly and finds most periods, with minor errors or limited justification.	Gives basic classifications . Period calculation or justification is incomplete and needs guidance.	Unable to classify the signals correctly or justify the classification .	**25**	**2.5**	

		phase, and growth or decay. Explains the reasoning clearly with mathematical conditions and waveform evidence.						
4. Analyze the amplitude and time-index characteristics of discrete-time signals.	**Amplitude & Time-Index Analysis**	Correctly determines maximum and minimum amplitude, the index n at which each occurs, and signal energy. Correctly shows the effect of time shifting and folding on the index. Interprets the results clearly.	Analyzes most amplitude and index characteristics correctly, with minor errors in values or interpretation.	Gives basic amplitude and index analysis with noticeable errors, and needs guidance.	Unable to analyze the amplitude and time-index characteristics or interpret the results.	**25**	**2.5**	
					Total	**100**	**10**	